BUSINESS CASE

DATA WAREHOUSING PROJECT
DATA WAREHOUSING TEAM
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COURSE: COSC471
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1. Executive Summary

This business case outlines the deployment of Data Warehouse (DW) architecture designed to transfer real-time stock information from a Data Collection team to an XGBoost system, designed to predict future stock values. The project delivers a fully automated, high-performance pipeline that ingests, validates, and engineers financial features, replacing legacy hardware with scalable supercomputing infrastructure available on the Digital Research Alliance of Canada (DRAC) cluster.

1.1. Issue

The XGBoost project requires a remotely accessible data warehouse service, that ingests real-time market data from the Data Collection team's database, creates features, and formats the data to be used for training in a XGBoost machine learning model. The data warehouse system must be hosted on a remote server, with high system security, to prevent historical data loss. Previous iterations relying on local college (Okanagan College, Kelowna) lab environments presented severe bottlenecks regarding network reliability, storage limitations, and hardware-level risks.

1.2. Anticipated Outcomes

The project will deliver a centralized, timing-based ingestion pipeline natively hosted on the DRAC High-Performance Computing (HPC) cluster. The end state is a fully automated system that validates OHLCV (Open, High, Low, Close, Volume) data and stores it in a PostgreSQL 16 database. By executing transformations directly in user-space via optimized SQL procedures, this architecture provides the XGBoost system with a "source of truth" for training models to predict future stock values.

1.3. Justification

This project is essential for environment isolation (Development, Test, and Production) and data integrity. By pivoting from local school hardware to the DRAC supercomputing infrastructure, the project mitigates the risk of total data loss in the school lab environment. Additionally, the implementation of a three-tier Data Engineering Quality Assurance (QA) testing suite mathematically guarantees the integrity of the data before it reaches the downstream predictive models.

2. Business Case Analysis Team

The following individuals comprise the business case analysis team. They are responsible for the analysis and creation of the Data Warehousing Project business case.

Role	Description	Name/Title
Product Owner	Handles client meetings, as well as interaction and project requirements as needed by the client.	Alex Anthony
Scrum Master	Manages sprints and keeps track of sprint information and kanban board issues.	Hayden Nikkel
Programmer	Completes project requirements	Jackson Rosco
Programmer	Completes project requirements	Daemon Lewis
Programmer	Completes project requirements	John Cortez

3. Project Overview

3.1. Project Description

The project involves a database-centric pipeline utilizing Bash scripting and native PostgreSQL 16 stored procedures. Operating directly within the user-space of the DRAC HPC cluster, the system executes high-performance SQL transformations to ingest, validate, and engineer market data. It organizes this data into a Star Schema (Fact/Dimension tables), providing an optimized, query-ready foundation of financial data without the latency of intermediate API layers.

3.2. Goals and Objectives

- Reliable Ingestion: Ingest and store market data reliably from external sources.
- Data Transformation: Transform raw data into calculated features for ML queries.
- Security: Maintain secure remote access via OpenVPN and SSH keys.

3.3. Project Assumptions

The following preliminary assumptions have been identified for the Data Warehousing Project to ensure successful delivery and operational stability:

- **Computer Availability:** The project assumes consistent uptime and job-scheduling availability within the DRAC HPC environment.

- **Storage Quotas:** It is assumed that the data warehouse will remain within the allocated user-space storage limits provided by the cluster, and that these academic quotas will not be unexpectedly reduced.
- **Data Consistency:** It is assumed that the Data Collection team will provide a consistent "source of truth" for OHLCV data that conforms to the expected schemas processed by the staging environment.
- **Training and Competency:** All project staff and maintainers are assumed to be trained in their respective tasks, particularly regarding PostgreSQL database administration, SQL optimization, and Bash scripting within an HPC environment.
- **Funding Availability:** It is assumed that necessary funding is secured for the procurement of required hardware (such as the Raspberry Pi 5 and 2TB storage) and any software licensing.
- **Infrastructure Support:** It is assumed that department heads and lab technicians will provide the necessary network and infrastructure support during development.
- **Executive Backing:** The project operates under the assumption of continued executive-level support and approval from the project sponsors and college faculty.

4. Strategic Alignment

Goals/Objectives	Relationship to Project
Environment Isolation	Schema separation (staging, dw, star, ml) and transactional black-box testing ensure that pipeline logic can be developed and tested without ever threatening the stability of the production database.
Enterprise Reliability & Data Integrity	By leveraging DRAC's infrastructure, hardware failure risks are drastically reduced. Additionally, the automated QA test suite physically blocks mathematically impossible or orphaned data from ever entering the warehouse.
ML Readiness	Time-series features (rolling SMAs, price lags, overnight gaps) are engineered natively within PostgreSQL. The pipeline enforces strict data "warmup" rules, providing the downstread XGBoost models with a formatted feature matrix without null-poisoning.

5. Cost Benefit Analysis / Alternatives Analysis

- Status Quo: Doing nothing results in a high risk of data loss, relies on local college lab networks, and lacks the structured formatting required for XGBoost training.
- Bare-Metal HPC Migration (Selected): By migrating to the DRAC cluster, the project secured enterprise-grade compute power and storage. This eliminates physical hardware procurement and maintenance while providing exceptional processing speed and industrial-grade data security.
- Commercial Cloud Deployment (Alternative): Hosting the data warehouse on cloud computing was considered but ultimately rejected due to the prohibitive recurring monthly costs for resources.
- Local Hardware Development (Alternative): The original plan to build a custom off-site backup network utilizing Raspberry Pi hardware and OpenVPN tunnels was rejected due to the high single-points-of-failure and lab network latency.